

The ROS 2 baseline at three tiers: the submitted model, the same model on board-profiled costs, the board

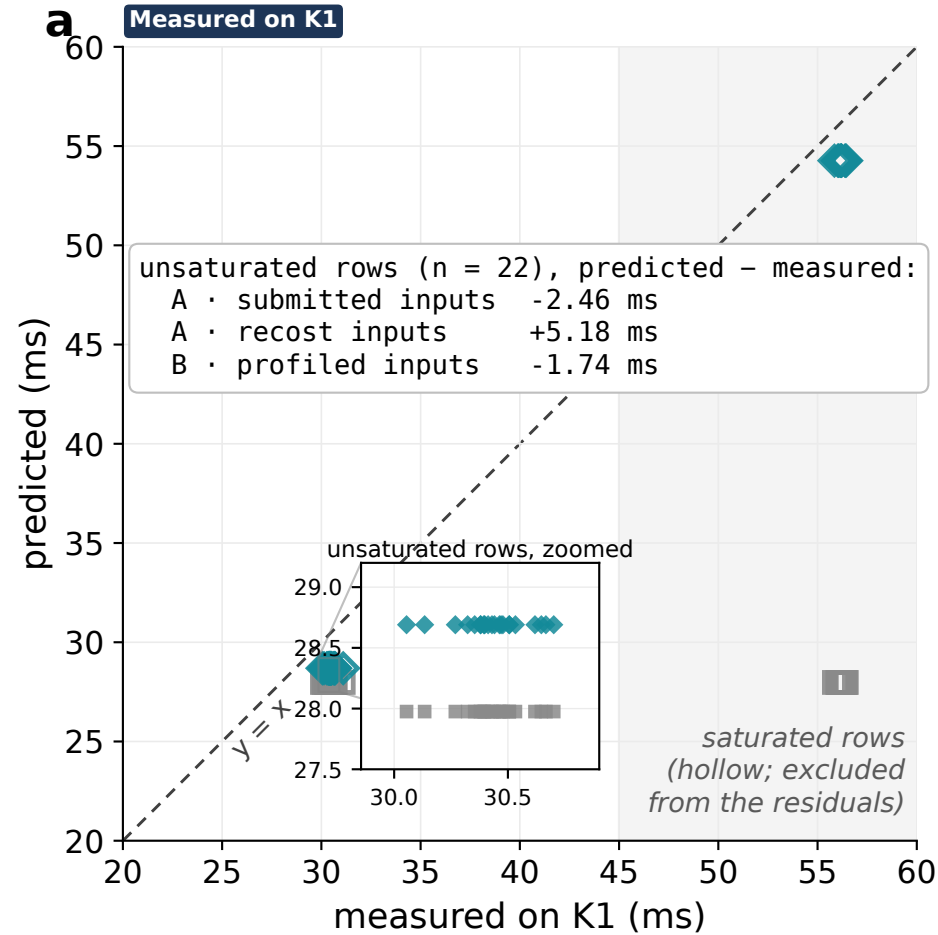
deployed camera → perception → nav → control chain · 35 (arm, camera rate) rows over 8 pinned arms, 5–120 Hz, 67 board runs

camera → goal latency

Analytical model

Calibrated model · board-profiled costs

Measured on K1

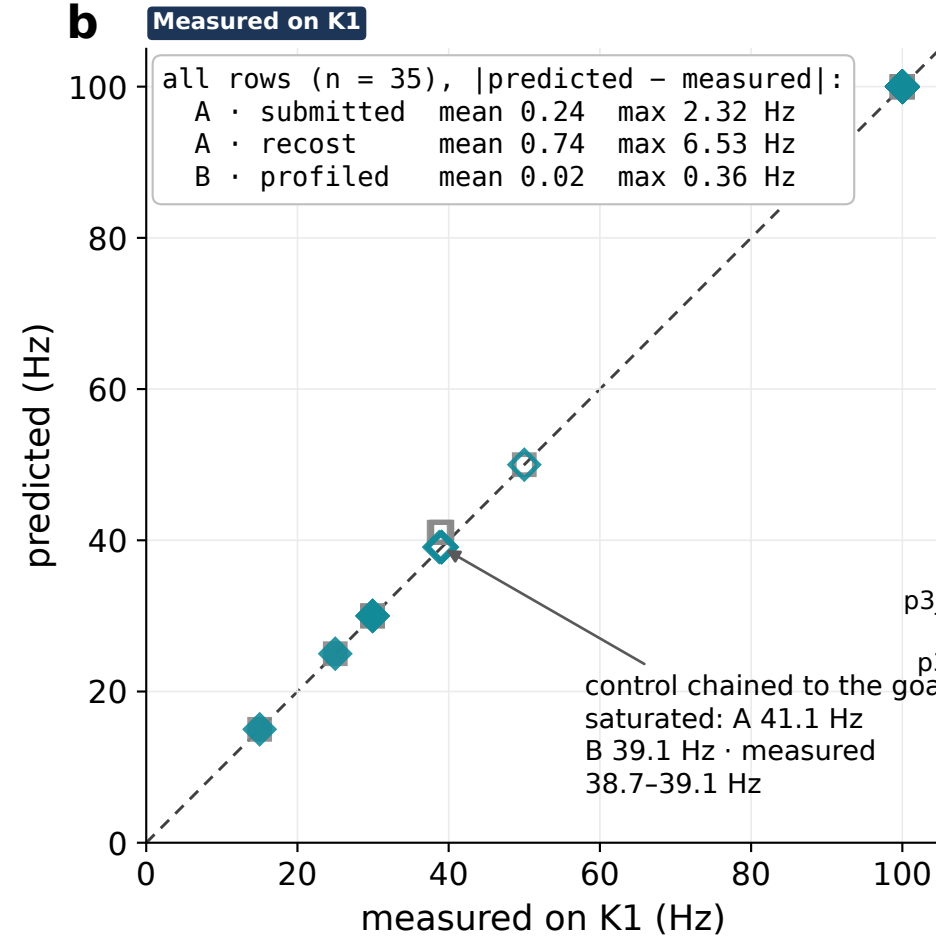


control command rate

Analytical model

Calibrated model · board-profiled costs

Measured on K1

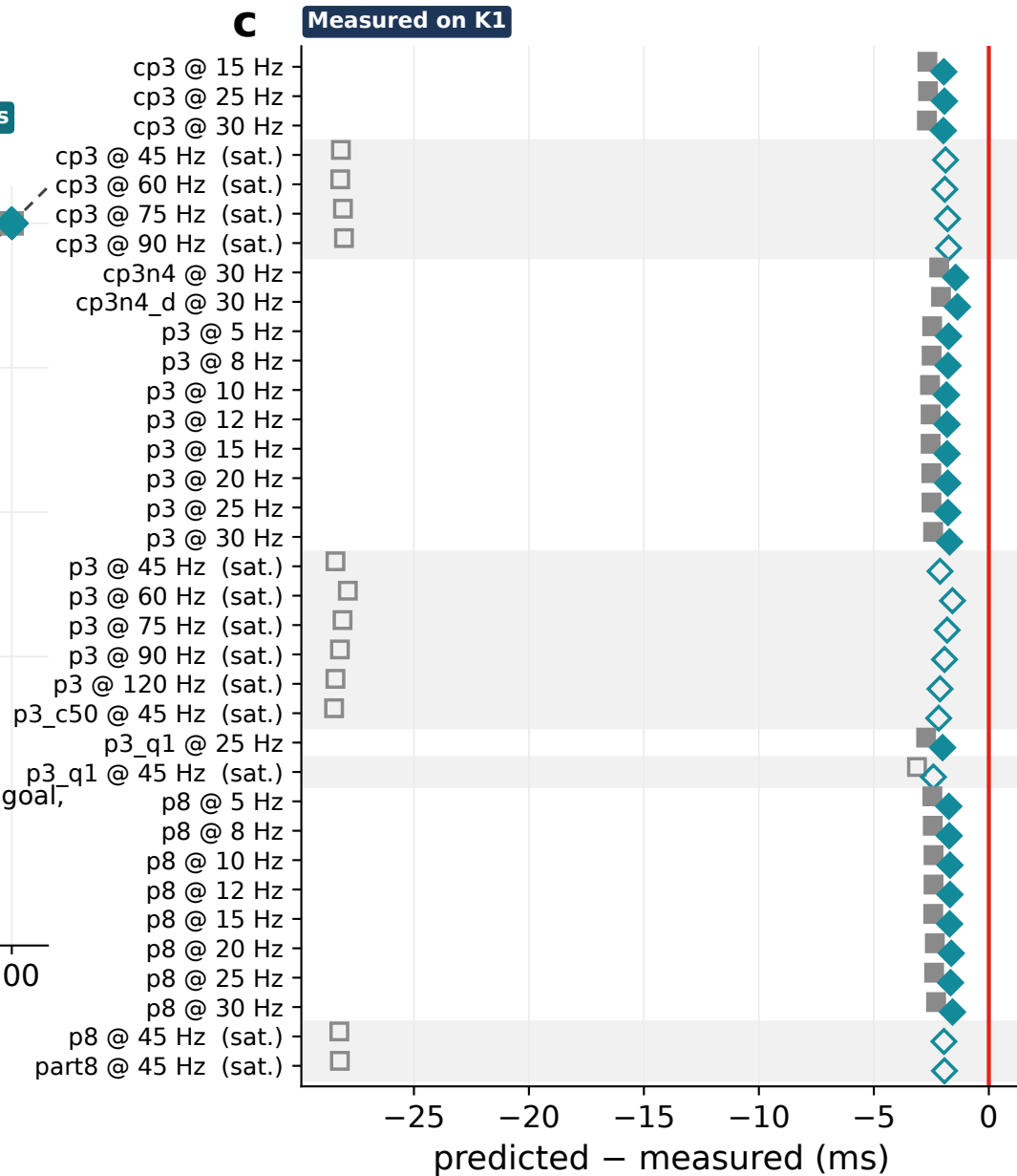


camera → goal error per row

Analytical model

Calibrated model · board-profiled costs

Measured on K1



■ A · analytical model, submitted inputs (XPU-RT's per-dispatch costs)

◆ B · same recurrence, per-node costs from the board's ROS 2 traces (+ named queue term)

— C · measured on K1 (x axis, or the 0 line)

○ hollow = saturated row